

Notes for ‘Polynomial and power functions’

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**Important Ideas and Useful Facts:**

- (i) **Power and  $n$ th root functions:** A *power function* has the rule  $y = x^n$  for some fixed exponent  $n$ . In the case that  $n$  is a positive integer, the value of  $x^n$  is  $x$  multiplied by itself using  $n$  occurrences of  $x$ . For example,

$$2^7 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$$

and

$$(-2)^7 = (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2) = -128.$$

When  $n = 1$  then  $y = x^1 = x$  becomes the *identity function*.

When  $n = 2$  then  $y = x^2$ , known as the *square function* that finds the area of a square of side length  $x$ .

When  $n = 3$  then  $y = x^3$ , known as the *cube function* that finds the volume of a cube of side length  $x$ .

- (ii) **Fractional powers and  $n$ th roots:** If we “undo” the function  $y = x^n$  (using the formal notion of an *inverse function*, discussed later) then we get the  *$n$ th root function*, written equivalently as a *fractional power*,

$$y = \sqrt[n]{x} = x^{1/n}.$$

When  $n = 2$  this becomes the usual *square root function*  $y = \sqrt{x} = x^{1/2}$ , which “undoes” the square function.

When  $n = 3$  this becomes the usual *cube root function*  $y = \sqrt[3]{x} = x^{1/3}$ , which “undoes” the cube function.

For example,

$$\sqrt{2} = 2^{1/2}, \quad \sqrt{3} = 3^{1/2}, \quad 2 = 4^{1/2} = 8^{1/3}, \quad 3 = 9^{1/2} = 27^{1/3}.$$

If  $a, b > 0$  then we have

$$\sqrt{ab} = \sqrt{a}\sqrt{b} \quad \text{and} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}},$$

both of which can be expressed in terms of fractional powers:

$$(ab)^{1/2} = a^{1/2}b^{1/2} \quad \text{and} \quad \left(\frac{a}{b}\right)^{1/2} = \frac{a^{1/2}}{b^{1/2}}.$$

For example,

$$8^{1/2}2^{1/2} = (8 \times 2)^{1/2} = 16^{1/2} = 4, \quad \frac{27^{1/2}}{3^{1/2}} = \left(\frac{27}{3}\right)^{1/2} = 9^{1/2} = 3.$$

More generally, if  $a, b > 0$  and  $n$  is any positive integer then

$$(ab)^{1/n} = a^{1/n}b^{1/n} \quad \text{and} \quad \left(\frac{a}{b}\right)^{1/n} = \frac{a^{1/n}}{b^{1/n}}.$$

(iii) **Polynomial functions:** A *polynomial function* (or simply *polynomial*) has the form

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

where  $n$  is a nonnegative integer and  $a_0, \dots, a_n$  are constants, with  $a_n \neq 0$ . We call  $n$  the *degree* of the polynomial or polynomial function.

When  $n = 0$ , we get a *constant function*  $p(x) = a_0$ .

When  $n = 1$ , we get a *linear function*  $p(x) = a_0 + a_1x$ .

When  $n = 2$ , we get a *quadratic function*  $p(x) = a_0 + a_1x + a_2x^2$ .

When  $n = 3$ , we get a *cubic function*  $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ .

A *root* of  $p(x)$  is a (real) number  $\lambda$  such that  $p(\lambda) = 0$ , where  $p(\lambda)$  is evaluated in  $\mathbb{R}$  using addition and multiplication. If  $\lambda$  is a root of  $p(x)$  then we have a factorisation

$$p(x) = (x - \lambda)q(x)$$

where  $q(x)$  is some polynomial of degree  $n - 1$ .

For example, let  $p(x) = x^2 - 3x + 2$ , a polynomial of degree 2. Observe that  $p(1) = 0$ , so that  $x - 1$  is a factor of  $p(x)$ . In fact,

$$p(x) = (x - 1)(x - 2),$$

so that  $p(x) = (x - 1)q(x)$  where  $q(x) = x - 2$ , a polynomial of degree 1.

(iv) **Division of polynomials:** If  $p(x)$  and  $s(x)$  are polynomials, we can *divide*  $p(x)$  by  $s(x)$  to get an equation

$$p(x) = s(x)q(x) + r(x)$$

for some polynomials  $q(x)$ , called the *quotient*, and  $r(x)$ , called the *remainder*, where either  $r(x) = 0$  or the degree of  $r(x)$  is less than the degree of  $s(x)$ . In the case where  $r(x) = 0$  then the previous equation becomes a factorisation  $p(x) = s(x)q(x)$ .

The process of finding the quotient  $q(x)$  and remainder  $r(x)$  is called *long division of polynomials* and mimics the usual long division of integers, by systematically “scooping out” copies of  $s(x)$  to methodically reduce the degree of the polynomial, and is illustrated in the examples below.

### Examples and derivations:

1. Let  $p(x) = x^2 - 3x + 2$ , the polynomial of degree  $n = 2$  discussed above. We noted that  $p(1) = 0$ , so that we are sure that  $s(x) = x - 1$  is a factor of  $p(x)$ .

We want to illustrate a general method for finding a polynomial  $q(x)$  of degree  $n - 1$  such that  $p(x) = s(x)q(x)$ . We know in advance, in this case, by inspection, that  $q(x) = x - 2$ , but want to get the same answer using the method of long division of polynomials.

To start, we set up a division symbol, with the polynomial  $p(x) = x^2 - 3x + 2$  that is being divided inside the symbol, and the polynomial  $s(x) = x - 1$  used to form the division outside (to the left of) the division symbol.

$$x - 1 \overline{) x^2 - 3x + 2}$$

The aim is to “scoop out” copies of  $s(x)$  from  $p(x)$  to successively remove the highest powers of  $x$  that appear inside the division symbol. Presently, the highest power of  $x$  is  $x^2$ , so we use  $x$  “copies” of  $s(x)$ , as this will produce an  $x^2$  term.

We place  $x$  on top of the division symbol, and  $x$  copies of  $x - 1$ , which is  $x(x - 1) = x^2 - x$ , below:

$$\begin{array}{r} x \\ x-1 \overline{) \begin{array}{r} x^2 - 3x + 2 \\ x^2 - x \end{array}} \end{array}$$

We then rule a line underneath and take  $x^2 - x$  away from  $x^2 - 3x + 2$ , which has the effect of making the  $x^2$  term go away, leaving the linear term  $-2x + 2$ :

$$\begin{array}{r} x \\ x-1 \overline{) \begin{array}{r} x^2 - 3x + 2 \\ x^2 - x \\ \hline -2x + 2 \end{array}} \end{array}$$

We wish now to make the  $-2x$  term go away, so we “scoop out”  $-2$  copies of  $x - 1$ , placing  $-2$  on top, and  $(-2)(x - 1) = -2x + 2$  below:

$$\begin{array}{r} x-2 \\ x-1 \overline{) \begin{array}{r} x^2 - 3x + 2 \\ x^2 - x \\ \hline -2x + 2 \\ -2x + 2 \end{array}} \end{array}$$

We rule another line underneath and take  $-2x + 2$  away, leaving 0:

$$\begin{array}{r} x-2 \\ x-1 \overline{) \begin{array}{r} x^2 - 3x + 2 \\ x^2 - x \\ \hline -2x + 2 \\ -2x + 2 \\ \hline 0 \end{array}} \end{array}$$

This shows that  $p(x) = x^2 - 3x + 2$  is really  $x - 2$  copies of  $x - 1$ , that is,

$$p(x) = (x - 1)(x - 2) ,$$

and so we have achieved the required factorisation, where we may take  $q(x) = x - 2$ , the linear polynomial appearing on top of the division symbol.

2. For a more difficult example, that can't easily be solved by inspection, consider

$$p(x) = x^3 + 3x^2 - 13x - 15 .$$

Observe that

$$p(1) = -24, \quad p(2) = -21 \quad \text{and} \quad p(3) = 0 .$$

We have stumbled on a root of  $p(x)$ , namely 3, so we know that  $x - 3$  is a factor of  $p(x)$ . Thus  $p(x) = (x - 3)q(x)$  for some polynomial  $q(x)$  of degree one less than the degree of  $p(x)$ , so that  $q(x)$  will be a quadratic.

We will find  $q(x)$  by long division of polynomials. First write down a division symbol, with  $p(x)$  inside and  $x - 3$  outside:

$$x - 3 \overline{) x^3 + 3x^2 - 13x - 15}$$

The aim is to “scoop out” copies of  $x - 3$  to systematically reduce the degree of the polynomials inside the division symbol. To start, we want to make the  $x^3$  term go away, so we use  $x^2$  copies of  $x - 3$ , writing  $x^2$  on top, and  $x^2(x - 3) = x^3 - 3x^2$  below:

$$x - 3 \overline{) x^3 + 3x^2 - 13x - 15} \\ x^3 - 3x^2$$

We then rule a line underneath and take  $x^3 - 3x^2$  away from  $x^3 + 3x^2 - 13x - 15$ , which has the effect of making the  $x^3$  term go away, leaving the quadratic  $6x^2 - 13x - 15$ :

$$x - 3 \overline{) x^3 + 3x^2 - 13x - 15} \\ x^3 - 3x^2 \\ \hline 6x^2 - 13x - 15$$

We wish now to make the  $6x^2$  term go away, so we “scoop out”  $6x$  copies of  $x - 3$ , placing  $6x$  on top, and  $6x(x - 3) = 6x^2 - 18x$  below:

$$x - 3 \overline{) x^3 + 3x^2 - 13x - 15} \\ x^3 - 3x^2 \\ \hline 6x^2 - 13x - 15 \\ 6x^2 - 18x$$

We rule another line underneath and take  $6x^2 - 18x$  away, leaving  $5x - 15$ :

$$x - 3 \overline{) x^3 + 3x^2 - 13x - 15} \\ x^3 - 3x^2 \\ \hline 6x^2 - 13x - 15 \\ 6x^2 - 18x \\ \hline 5x - 15$$

We wish now to make the  $5x$  term go away, so we “scoop out” 5 copies of  $x - 3$ , placing 5 on top, and  $5(x - 3) = 5x - 15$  below:

$$\begin{array}{r}
 x^2 + 6x + 5 \\
 \hline
 x - 3 \overline{) x^3 + 3x^2 - 13x - 15} \\
 \underline{x^3 - 3x^2} \phantom{- 13x - 15} \\
 6x^2 - 13x - 15 \\
 \underline{6x^2 - 18x} \phantom{- 15} \\
 5x - 15 \\
 \underline{5x - 15} \\
 0
 \end{array}$$

We rule another line underneath and take  $5x - 15$  away, leaving 0:

$$\begin{array}{r}
 x^2 + 6x + 5 \\
 \hline
 x - 3 \overline{) x^3 + 3x^2 - 13x - 15} \\
 \underline{x^3 - 3x^2} \phantom{- 13x - 15} \\
 6x^2 - 13x - 15 \\
 \underline{6x^2 - 18x} \phantom{- 15} \\
 5x - 15 \\
 \underline{5x - 15} \\
 0
 \end{array}$$

This shows that  $p(x) = x^3 + 3x^2 - 13x - 15$  is really  $x^2 + 6x + 5$  copies of  $x - 3$ , that is,

$$p(x) = (x - 3)(x^2 + 6x + 5) ,$$

and so we have achieved the required factorisation, where we may take  $q(x) = x^2 + 6x + 5$ , the quadratic polynomial appearing on top of the division symbol.

Being a quadratic, we can factorise  $q(x)$  directly, in fact by inspection in this case (though if that were difficult we could rely on the quadratic formula to find the roots and therefore the linear factors), to get

$$p(x) = (x - 3)(x^2 + 6x + 5) = (x - 3)(x + 1)(x + 5) ,$$

achieving a complete factorisation of the cubic  $p(x)$ .

If, for example, we need to solve the equation  $p(x) = 0$ , then we could use this factorisation to deduce immediately that  $x - 3 = 0$ ,  $x + 1 = 0$ , or  $x + 5 = 0$ , yielding solutions  $x = 3, -1, -5$ .

3. Suppose that  $\lambda$  is a root of a polynomial  $p(x)$  of degree  $n$ . We explain briefly why it must be the case that

$$p(x) = (x - \lambda)q(x)$$

for some polynomial  $q(x)$  of degree  $n - 1$ .

*Proof:* We can divide  $x - \lambda$  into  $p(x)$ , using the method of the previous two examples, producing finally a constant remainder, by systematically “scooping out” copies of  $x - \lambda$  to make all of the terms under the long division symbol involving  $x^n, x^{n-1}, \dots, x^2, x$  successively go away.

Denote by  $r$  the final remainder, which is a constant, that appears at the end of the division, and let  $q(x)$  be the polynomial that finally appears on the top of the division symbol. The long division tells us that

$$p(x) = (x - \lambda)q(x) + r .$$

But  $\lambda$  is a root of  $p(x)$ , so that, substituting  $\lambda$  for  $x$  in the previous equation, we get

$$0 = p(\lambda) = (\lambda - \lambda)q(\lambda) + r = 0q(\lambda) + r = 0 + r = r .$$

This shows that the constant remainder  $r$  is in fact zero, so that

$$p(x) = (x - \lambda)q(x) ,$$

completing the proof.